Supplementary Information: Experiment-Driven Bayesian Optimization of the Impact Response of Multi-Material 3D-Printed Tensegrity-Inspired Structures

This Supplementary Information collects the detailed design-, fabrication-, test-, and analysis-development records supporting the main manuscript. Unlike the main text, the SI references the project's development history directly, including specific pull requests (PRs) and issues in the project repository (https://github.com/vertical-cloud-lab/tensegrity-optimization).

S1 Joint-Design Study

The strut-to-cable junction was developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants). The working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies; a captive TPU core is routed inside the PLA shell so that the soft tendon is protected from layer-line failure at the transition.

Full results, renders, and the Phase-3/Phase-4 design reviews are documented in PR #39 (joint design) and PR #35 (CAD parameterization and print history).

Placeholder: joint-design variants and the selected dovetail/anchor-bulb junction (import figures from PR #39 / PR #35).

Figure S1: Five candidate strut-to-cable joint geometries and the captive-TPU-core dovetail junction selected for the working prototype.

S2 Support Strategy for Soft Members

Near-vertical TPU tendons are otherwise unsupported during printing. For the printed campaign batches, automatic support generation was disabled and PLA edge supports were painted manually in Bambu Studio along each tendon, with the support-to-object XY clearance tightened from 0.35 mm to 0.2 mm during tuning on the S0 reference prism (issue #98). Support removal is the leading source of the logged cosmetic defects (surface stringing; on two seed articles, TPU strands pulled off bottom tendons), so defects are recorded per article in the print key (Section S3).

An automated alternative was developed in parallel: a tensegrity-specific Bambu Studio recipe (support threshold angle lowered from 40° to 10°, support material matched to the rigid extruder) combined with manually generated narrowing pillars that ray-cast against the printable mesh underside. The narrowing-pillar workflow is documented in PR #66; the H2D multi-part material assignment fix is in PR #64.

S3 As-Printed Campaign Record (Print Key)

Table S1 is the print key for the seed campaign: every article's print identifier, its design, its weighed mass, and its logged defects. The authoritative machine-readable key, together with the as-printed slicer project (one specimen per named build plate), lives at PR #102 (`bo/t3-prism-bo-batch-print-key.csv`). Two entries deserve care. Design 08 was printed three times; the plate label in the archived slicer project and a contemporaneous issue comment disagree on which article is the official one (*dea4ls* versus *bag26v*), and pending confirmation the analysis uses *bag26v*. The tri-print of design 08 also supplies the only print-to-print repeatability number to date: a mass standard deviation of 0.457 g (three prints of a single geometry, so not a general process repeatability estimate). Separately, one tested article (*amdjwm*) has no confirmed design mapping and is excluded from the surrogate fit until identified.

Table S1: Print key for the seed campaign (nine Sobol designs plus the S0 reference). Masses weighed on a 0.01 g scale. TPU fed from a heated dry box near 8% relative humidity where logged. The first five articles (printed before the TPU nozzle fix) used the standard 0.4 mm TPU nozzle; the rest used the high-flow 0.6 mm nozzle.

Design	Article	Mass (g)	Logged defects
Spec 00	9hhbkp	21.62	none
Spec 01	6lhxfy	18.50	slight strut misprint; divot on one top tendon
Spec 02	autv5r	22.04	minor tendon misalignment near top
Spec 03	ebdna8	19.61	two strands off bottom tendons (support removal)
Spec 04	nvxsrv	20.66	misalignment ring after printer pause
Spec 05	6nheas	21.73	two strands off bottom tendons (support-related)
Spec 06	1zm8rv	18.50	stringing along diagonal tendons
Spec 07	ajhby6	20.53	filament line pulled from bottom struts
Spec 08	dea4ls	22.29	stringing; residual PLA on tendon
Spec 08	bag26v	21.42	some stringing on tendon
Spec 08	ghmj4y	22.10	inconsistent tendon cross-section near top
S0 ref.	bpx68c	20.23	none major

S4 Drop-Tower Protocol and Rig Characterization

The campaign fixture is a Lansmont Model 23 shock test system with a Test Partner 4 data-acquisition console. The development history behind the protocol in the main text is summarized here.

Protocol. 60 in drop height; specimen upright on the carriage plate; 101 valid captures per specimen (103 for one article, autv5r), counted after the first two drops of each session are discarded as warm-up; three slow-motion video clips (959 fps) per specimen; roughly 42 s per drop and about ninety minutes per specimen. Capture settings: 1.25 MHz sample rate, 100 ms record, 2 ms pre-trigger, 150 G trigger on the base channel, AC-coupled ICP conditioning. The console settings were re-verified digit-for-digit against an archived screenshot after a settings-loss event, with data continuity confirmed to within about 1%.

Impact surface. A blinded 3×3 crossover experiment over three candidate polyurethane mat arrangements (single 0.25 in Shore 70 OO; single 0.5 in Shore 60 OO; stacked) selected the single 0.5 in Shore 60 OO sheet (issue #88, PR #86). Unlike the earlier felt stack, the mat does not measurably compact over hundred-drop sessions. Historical note: sessions recorded as "5 felts" in early protocol documents were physically four felt sheets plus one cardboard layer.

Sensors and calibration. Base input: single-axis piezoelectric accelerometer on the carriage plate (1.059 mV/G, full scale 9443 G); top response: Dytran 3133A4 triaxial accelerometer (X/Y/Z 0.690/0.667/0.734 mV/G) seated in a printed pocket at the top vertex, mounted per ISO 5348. An earlier single-axis channel was retired after range clipping. The two base-capable channels were cross-calibrated in place over fifteen co-located drops: ratio 0.953 ± 0.001 at CFC-180 (per-drop 0.952 ± 0.004), stable across drop heights (PR #74).

Filtering correction. The SAE J211/1 channel-frequency-class filter used for all reported numbers is implemented from the standard's published coefficients. An earlier in-house implementation passed the two-pass (forward-backward) corner frequency to a single-pass design, which narrows every channel class by roughly 20% (an effective CFC-144 where CFC-180 was intended); the discrepancy was identified during the analysis audit of issue #94 and corrected before the campaign numbers were produced.

Processing pipeline. For each capture, every channel is calibrated and CFC-filtered independently; the top-vertex magnitude $\|\mathbf{a}_{\text{top}}(t)\|$ is formed from the three filtered triaxial components, and the two peaks entering t_{180} are searched within a short window (approximately 1.5 ms) of first contact, so the peaks need not be simultaneous. The baseline is estimated from the record tail (at healthy tower speed the mat-contact foot begins before the 2 ms pre-trigger window). Δv is the integral of the baselined base channel over the impact pulse and doubles as the rig-health gate. Drops are discarded per the warm-up rule; remaining per-capture exclusions, the second-contact detector's threshold and search window, the exact window bounds and filter-initialization choices, and the "stabilized mean" convention are defined in the campaign analysis code, whose archival release with a version tag is a declared submission gate; they are summarized here rather than restated numerically to avoid divergence from the code of record.

Rig health and repair history. Rig health is gated on the impact velocity change Δv recovered from the base channel; over an uninterrupted 100-drop session the Δv CV was 0.54% with a flat drift slope. During commissioning, two guide-rod locking pins failed and a defeated pin-park safety interlock was discovered and recommissioned; the pin-related friction produced a measurable Δv deficit that resolved after repair (issue #92). Approximately 3,660 instrumented drops across roughly 50 printed specimens were performed over three months of commissioning and characterization before the campaign sessions (issue #101); the campaign itself contributes over 800 clean captures to date.

Objective provenance and adversarial review. The objective stack evolved through a measurability review (PR #97): per-specimen crush energy in joules is not recoverable from this rig (no falling-mass or displacement channel, and the specimens stay elastic), which is why the campaign objectives are the filtered peak-acceleration ratio and the rebound energy, with quasi-static crush metrics deferred to a load-frame campaign. A pre-registered adversarial review of the objective choices (Edison ANALYSIS task `3e398131`, archived in the repository) subsequently recommended refitting with article-level noise (0.72% CV rather than per-drop standard errors), treating the rebound fraction as a candidate constraint pending a restrained-versus-unrestrained validation with synchronized video, and reserving print capacity for replicate articles; these corrections define the round-3 plan described in the main-text Discussion.

S5 Printed-Mass Model

The seed batch was projected onto constant *solid* CAD mass ($m^\star = 30.95$ g), yet the articles weigh 18.50 g to 22.29 g, because PLA struts print at partial infill (15% grid, two walls) while thin TPU cables print near solid, and the PLA-to-TPU volume split swings widely across the design box. A two-stage calibrated model closes this gap (PR #102, `bo/t3_prism_mass_model.py`): analytic member volumes are corrected to rendered-solid grams (one factor per material), and rendered-solid grams are mapped to weighed grams through a wall-plus-infill term in the as-printed strut diameter (fitted infill 23.6%, wall 1.18 mm, TPU factor 0.744). The model's residual standard deviation is 0.378 g. For context, the three available prints of design 08 have a sample standard deviation of 0.457 g; three prints of one geometry are insufficient to establish a general process repeatability limit. Inverting it inside the projection allows rounds to hold *printed* grams constant with no slicer in the loop. Two round-2 recommendation sets exist in the record and must not be conflated: the batch actually sliced and sent to the printer was generated before this correction, under the seed round's solid-mass projection, with the mass-aware objectives and each recommendation's predicted printed mass reported; a subsequently regenerated set applies the constant-printed-mass projection (target 20.23 g, the reference article's weight) and is the working proposal for round 3. The repository's suggestion CSV at the branch head is the regenerated (constant-printed-mass) set.

S6 Simulation Fidelity Ladder

The screening ladder of the main text was assembled and benchmarked in PR #33. Tier C is a MuJoCo rigid-strut, spring-tendon model (about 0.15 s per design with CFC-180 filtering); PyBullet and PyChrono equivalents were benchmarked and demoted (the PyBullet bare-prism peak is contact-dominated and design-invariant). Tier B is a Newton/Warp XPBD model with deformable struts and explicit tendon cross-sections (seconds to tens of seconds per design), which restores the tendon-stiffness sensitivity Tier C lacks. Tier A is a PolyFEM incremental-potential-contact hyperelastic model of the welded PLA/TPU prism (about a minute per run), which captures strut buckling and self-contact.

Documented blind spots carried into any use of the ladder: Tier-C peak acceleration is floor-contact dominated and nearly flat across cable stiffness; the apparent strut-diameter leverage in unconstrained sweeps is largely a mass/contact confound (it shrinks about seventeenfold under a constant-mass sweep); the twist variable is not consumed by the Tier-C regime plumbing; rigid-strut models cannot reproduce the measured amplification ($t_{180} > 1$ on five of eight bench articles, while simulated t_{180} cannot exceed about 1); and simulated restitution is design-invariant where the measured rebound is not. Against the measured seed round ($n = 7$ mapped articles), an intensive volumetric energy-absorption score computed at Tier C showed a strong exploratory in-sample rank association with measured t_{180} (Spearman $\rho = -0.93$, exact two-sided permutation $p \approx 0.0067$; hypothesis-generating rather than validated prediction, since the metric was evaluated on the seed set without multiplicity adjustment), while the ladder's own simulated t_{180} reaches only $\rho \approx 0.5$.

A simulation-side closed-loop study using the same Ax/BoTorch pipeline on the Tier-C evaluator (ten independent Sobol-seeded repeats of 36 designs each) reached 97.1% of an exhaustively computed reference-front hypervolume, versus 80.4% for compass search and 60 to 66% for random, Latin-hypercube, and Sobol baselines, with the smallest seed-to-seed spread; this motivates the budget-matched baseline comparison planned for the physical campaign but is not evidence about the physical loop itself.

S7 Metal-Analog Validation Plan

A final validation campaign rebuilds a subset of the T_3-prism designs as metal analogs (hollow aluminum rods for the compression members and stainless-steel threaded cables for the tension members) using the same R, H, θ, and member-diameter parameterization as the printed PLA/TPU specimens. To probe transfer across material systems rather than only confirming the optimum, the selected designs deliberately span the predicted performance range: representatives the surrogate predicts to be worst-, mid-, and best-performing are each built in both material systems and characterized under the identical drop protocol of Section S4. Agreement between the measured PLA/TPU ranking and the hollow-aluminum/stainless-steel ranking on the primary campaign metric t_{180} (reported in the main-text metal-analog validation table and figure, with rebound energy and, later, quasi-static SEA as secondary rankings) indicates that the optimization generalizes beyond the as-printed polymers.

Placeholder: assembled metal-analog specimens (hollow aluminum rods + stainless-steel threaded cables) beside their printed PLA/TPU equivalents for the worst/mid/best predicted tiers (photographs pending fabrication).

Figure S2: Metal-analog validation specimens alongside their printed PLA/TPU equivalents for each predicted performance tier.